# IRO – Issues Reporting Outlet: Object Listing

CSC426 Weekly Dev – Weekend 3 · Joshua Omolafe (Joshua989)

IRO is written in plain JavaScript as a small set of singleton objects. Each page has one controller object, and all of them share the configuration object `IRO` and the data-layer object `IssueStore`. The tables below list every object with its properties and methods.

## 1. Object summary

| Object | File | Kind | Used by | Purpose |
|---|---|---|---|---|
| `IRO` | js/store.js | Configuration | All pages | Constant lists and settings. |
| `IssueStore` | js/store.js | Data layer | All controllers | Create, read, update and delete issues; statistics. |
| Helper functions | js/store.js | Utilities | All controllers | Escaping, date formatting, badges, filling selects. |
| `ReportForm` | js/report.js | Controller | report.html | Report submission. |
| `Tracker` | js/track.js | Controller | track.html | Status tracking by ticket ID. |
| `AdminDesk` | js/admin.js | Controller | admin.html | Desk officer dashboard. |
| Issue | (data record) | Entity | IssueStore | One reported issue stored in localStorage. |

## 2. IRO (configuration object)

| Property | Type | Value / description |
|---|---|---|
| `STORAGE_KEY` | String | "iro_issues" – localStorage key holding all issues. |
| `ADMIN_PASSCODE` | String | "iro2026" – demo passcode for the admin desk. |
| `CATEGORIES` | Array<String> | 10 issue categories (Security / Safety … Other). |
| `COMMUNITIES` | Array<String> | Akungba, Iwaro, Etioro, Ayegunle, Oke-Oka, Other. |
| `STATUSES` | Array<String> | Submitted, Under Review, Escalated, Resolved, Closed. |
| `SEVERITIES` | Array<String> | Low, Medium, High, Emergency. |

## 3. IssueStore (data-layer object)

| Method | Returns | Description |
|---|---|---|
| `getAll()` | Array<Issue> | Reads all issues from localStorage, newest first; returns [] if data is missing or corrupt. |
| `saveAll(list)` | void | Writes the whole list back to localStorage as JSON. |
| `generateId()` | String | Builds a unique ticket ID IRO-YY-XXXXX from an unambiguous alphabet. |
| `create(data)` | Issue | Builds an Issue from form data (drops name/contact if anonymous), sets status Submitted with first history entry, saves it. |
| `findById(id)` | Issue \| null | Case-insensitive lookup by ticket ID. |
| `updateStatus(id, status, note)` | Issue \| null | Sets the new status and appends {status, note, at} to history. |
| `remove(id)` | void | Deletes an issue. |

| | | |
|---|---|---|
| `stats()` | Object | { total, open, emergency, byStatus } for the dashboard. |

### 4. Helper functions (global)

| Function | Returns | Description |
|---|---|---|
| `escapeHtml(str)` | String | Replaces & < > " ' with HTML entities to prevent XSS. |
| `formatDate(iso)` | String | Formats an ISO date, e.g. "7 Jul 2026, 14:05". |
| `statusBadge(status)` | String (HTML) | Coloured badge markup for a status. |
| `fillSelect(select, items, placeholder)` | void | Fills a <select> with option elements. |

### 5. ReportForm (controller for report.html)

| Member | Type | Description |
|---|---|---|
| `form` | HTMLFormElement | The #reportForm element. |
| `el(id)` | Function | Shortcut for document.getElementById. |
| `init()` | Method | Fills drop-downs, sets the max date, attaches event listeners. |
| `toggleReporterFields()` | Method | Shows matric field for students, community list for indigenes. |
| `toggleIdentity()` | Method | Hides name/contact when Anonymous is ticked. |
| `collect()` | Method → Object | Reads every form field into a plain object. |
| `validate(d)` | Method → String | Returns an error message, or "" if valid. |
| `submit(e)` | Method | Validates, calls IssueStore.create(), shows the ticket ID. |

### 6. Tracker (controller for track.html)

| Member | Type | Description |
|---|---|---|
| `el(id)` | Function | Shortcut for document.getElementById. |
| `init()` | Method | Attaches the form handler; auto-tracks an `?id=` in the URL. |
| `show(id)` | Method | Finds the issue and renders title, meta, status and history (no personal data). |

### 7. AdminDesk (controller for admin.html)

| Member | Type | Description |
|---|---|---|
| `SESSION_KEY` | String | "iro_admin" – sessionStorage flag for a logged-in officer. |
| `selectedId` | String \| null | Ticket ID of the report open in the detail panel. |
| `el(id)` | Function | Shortcut for document.getElementById. |
| `init()` | Method | Fills filters, attaches listeners, restores an existing session. |
| `login(e)` / `logout()` | Method | Checks passcode and opens the dashboard / ends the session. |
| `showDashboard()` | Method | Hides login card, shows dashboard, renders data. |

| | | |
|---|---|---|
| `filtered()` | Method → Array | Issues matching search text and status/category/reporter filters. |
| `render()` | Method | Updates statistics cards and the report table. |
| `openDetail(id)` | Method | Shows full report details, reporter identity and history. |
| `saveUpdate(e)` | Method | Calls IssueStore.updateStatus() with the chosen status and note. |
| `deleteSelected()` | Method | Confirms, then deletes the open report. |

### 8. Issue (data record)

| Property | Type | Description |
|---|---|---|
| `id` | String | Ticket ID. |
| `reporterType` | String | Student \| Indigene \| Other. |
| `anonymous` | Boolean | Reporter chose anonymity. |
| `name`, `contact` | String | Empty when anonymous. |
| `matricNo`, `community` | String | Student matric number / indigene community. |
| `category`, `severity` | String | From IRO.CATEGORIES / IRO.SEVERITIES. |
| `location`, `incidentDate` | String | Where and when (YYYY-MM-DD). |
| `title`, `description` | String | Summary and full account. |
| `status` | String | Current status. |
| `createdAt` | String (ISO) | Submission time. |
| `history` | Array<{status, note, at}> | Append-only status log. |

### 9. HTML page objects (main DOM elements)

| Page | Key element IDs |
|---|---|
| index.html | #categoryList |
| report.html | #reportForm, #reporterType, #matricGroup, #matricNo, #communityGroup, #community, #anonymous, #identityGroup, #name, #contact, #category, #severity, #location, #incidentDate, #title, #description, #formError, #formCard, #successCard, #ticketId, #trackLink |
| track.html | #trackForm, #ticketInput, #trackError, #resultCard, #resTitle, #resMeta, #resStatus, #resHistory |
| admin.html | #loginCard, #loginForm, #passcode, #loginError, #dashboard, #logoutBtn, #statTotal, #statOpen, #statEmergency, #statResolved, #search, #filterStatus, #filterCategory, #filterReporter, #issueRows, #emptyMsg, #detailCard, #dTitle, #dMeta, #dReporter, #dDesc, #dHistory, #updateForm, #newStatus, #note, #deleteBtn |